Radial Quadratic Extraction and Measure-Aware Certificates for the Bi-Orthogonally Invariant Spherical Perceptron

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

The source separates a spherical perceptron pressure into a spectral allocation entropy and a Haar angular pressure. Conditional on its angular limit, we extend this reduction to activations $f(z) = cz^2 + g(z)$, with $c \in \mathbb{R}$ and continuous subquadratic g, for $M = \lfloor \alpha N \rfloor$, by extracting the quadratic energy before the auxiliary Gaussian hierarchy. Thus positive quadratic growth never enters a Gaussian terminal class that excludes it. For bounded Lipschitz g, we derive an entropy-constrained spectral-block perturbation bound, evaluated by a one-dimensional multiplier equation with the actual block proportions. A mass-preserving interior regularization then gives an explicit radial-grid enclosure, including angular oracle errors and constants independent of the smallest positive block mass. Exact quadratic examples explain both the normalizations and why negligible spectral mass alone cannot remove an outlier. Classical entropy duality, Gaussian differentiation, and Dirichlet asymptotics are distinguished from these source-specific applications. Haar geometry, synchronization, the cavity lower bound, and the full variational equality remain upstream inputs rather than consequences of finite calculations. Contribution assessment: C3/5, a conditional quadratic-growth extension and explicit measure-aware spectral and radial approximation certificates provide a notable local increment without independently certifying the source's global variational theorem.

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1 The physical model and its actual measure

For a real $M \times N$ matrix A and uniform *probability* measure σ_N on $\{x : \|x\|^2 = N\}$, define

$$F_N(A, f) = \frac{1}{N} \log \int \exp\left\{ \sum_{r=1}^{M} f((Ax)_r) \right\} d\sigma_N(x). \tag{1}$$

The aspect ratio is $M/N \to \alpha \in (0, \infty)$. The source [1] assumes bi-orthogonal invariance and a deterministic compact limiting singular-slot law ν on $[0, \infty)$:

$$\frac{1}{\min(M, N)} \sum_r \delta_{s_{N,r}} \Rightarrow \nu, \qquad \max_r \operatorname{dist}(s_{N,r}, \operatorname{supp} \nu) \to 0 \tag{2}$$

in probability. These are rectangular singular slots, including zeros within them. Structural null input or output coordinates outside the slots are different objects. In particular forced left zeros cannot be used to excuse a small singular-slot outlier.

The source asserts a deterministic pressure limit for every bounded continuous activation. Its appendix also treats arbitrary dimension sequences, certain edge-interval spectral controls, continuous subquadratic activations, and bounded Borel activations when $\nu \neq \delta_0$. Those are existing source claims, not new extensions credited here. Our quadratic-growth extension below uses the stronger hypothesis (2) and the exact convention $M = \lfloor \alpha N \rfloor$. In particular $M = N$ at $\alpha = 1$. The broader dimension statement above is an imported bounded-activation result only; we do not transfer it to quadratic growth.

1.1 Finite blocks and radial entropy

Initially fix positive singular representatives $s_1, \ldots, s_h$ with dimensions $n_j/M \to p_j > 0$. Set $v_j = \alpha p_j$. Zero singular-slot blocks are also permitted: replace each zero representative by $\varepsilon > 0$, keep its dimension mass, and then let $\varepsilon \downarrow 0$. The operator comparison proved below makes this continuation unique. Its angular radius becomes zero, with the uniform measure on that radius-zero sphere interpreted as a point mass. This does not identify a zero singular-slot block with the structural input-null block. If $\alpha < 1$, add a structural input-null block with $v_0 = 1 - \alpha$ and $s_0 = 0$. Otherwise omit index zero. The index set I has $\sum_{i \in I} v_i = 1$. Write $\rho_i > 0$, $\sum_i \rho_i = 1$, for fractions of the input squared norm, not for dimension proportions. The output radii satisfy

$$\|y_j\|^2 = M a_j = N s_j^2 \rho_j, \qquad a_j = s_j^2 \rho_j / \alpha. \tag{3}$$

For mutually orthogonal Haar subspaces $E_j \subset \mathbb{R}^M$ of dimensions n_j, the angular pressure is

$$P_M^g(a) = \frac{1}{M} \log \int \exp\{\textstyle\sum_r g(\sum_j y_{j,r})\} \prod_j d\sigma_{E_j, M a_j}(y_j).$$

Directions are integrated at fixed subspaces. The source's angular theorem supplies a deterministic continuous limit $P_p^g(a)$, uniformly on compact radius sets in probability.

The spherical input gives a Dirichlet law with parameters $n_i/2$, so its logarithmic radial rate is

$$I_v(\rho) = \frac{1}{2} \sum_i v_i \log \frac{\rho_i}{v_i} = -\frac{1}{2} D(v \| \rho), \qquad D(v \| \rho) = \sum_i v_i \log \frac{v_i}{\rho_i}. \tag{4}$$

This is reverse relative entropy in the allocation variable. It is not $D(\rho \| v)$ and not an unweighted sum. Stirling's formula gives the rate directly from the Dirichlet density. It tends to $-\infty$ at a boundary coordinate with $v_i > 0$. All logarithms in this report are natural.

1.2 The full source route

The 1,465-line scientific manuscript was read in full. Its dependency structure is substantially more than a radial Laplace calculation.

Source lines	Mechanism and retained obligation
115–230	Powered outer/inner Stiefel integral, a first coordinate shared at level θ, matrix paths, active-space determinants, and successive witness limits.
232–353	Marked cascades, deletion of zero marks, independent residual noise on each visit, joint GG identities, synchronization, and matrix path variation.
355–520	Canonical Gaussian multiplier, strict precisions, compression of an inverse at freezing, matrix entropy, and moment control.
522–822	Enriched contact upper bound; two-pattern restoration, thick-shell cavity comparison, logarithmic integrability, and telescoping lower bound. These are not established by our finite tests.
824–1066	Quadratic tails, global soft constraints, Haar flags, weighted shell majorization, triangular entropy, and ordered limits $\theta \downarrow 0$, then cutoff, then penalty.
1068–1304	Dirichlet allocation, spectral approximation, scope extensions, and counterexamples to unrestricted spectral or activation interpretations.

The source's Gaussian formula uses terminals bounded above with at most quadratic negative growth. The sharing exponent remains positive; it is not an analytic continuation in replica number. Its strict active precisions and independently resampled visit residuals must remain intact.

2 Extracting quadratic growth before Gaussian reduction

Theorem 2.1 (Conditional quadratic-plus-subquadratic extension)**.** *Assume the source angular limit for bounded continuous activations, $M = \lfloor \alpha N \rfloor$, and the spectral hypotheses (2). Fix $c \in \mathbb{R}$ and a continuous g with $|g(z)|/(1+z^2) \to 0$ as $|z| \to \infty$. Then (1) for $f(z) = cz^2 + g(z)$ converges in probability to a deterministic limit. For bounded g and a finite spectral approximation its value is*

$$\mathcal{F}_{v,s}^{c,g} = \sup_{\rho \in \Delta_I^\circ} \left\{ I_v(\rho) + d \cdot \rho + \alpha P_p^g((s_j^2 \rho_j/\alpha)_j) \right\}, \qquad d_i = c s_i^2. \tag{5}$$

For general subquadratic g, take a final bounded continuous cutoff limit. The spectral mesh limit exists and is independent of its approximations. Convergence in mean holds if $\|A_N\|_{\mathrm{op}}^2$ is uniformly integrable; a deterministic uniform bound suffices.

Proof. For fixed singular values and input allocations, orthogonality gives

$$\sum_r (Ax)_r^2 = \|Ax\|^2 = N \sum_i s_i^2 \rho_i.$$

The quadratic weight is constant on each product of angular spheres. The remaining angular problem is exactly that for g, and the Dirichlet Laplace argument gives (5). Here $M = \lfloor \alpha N \rfloor$ is essential: for $\alpha < 1$ the structural input-null block has positive limiting mass $1 - \alpha$; for $\alpha \geq 1$ it is absent for all sufficiently large N. Thus this step omits no zero-mass structural input-null allocation. Because d and g are bounded at a fixed finite discretization, the divergent negative entropy excludes radial boundary points. Compact-interior angular convergence supplies both Laplace bounds.

For bounded Lipschitz g, any matrices of the same shape satisfy

$$\begin{aligned} |F_N(A, cz^2 + g) - F_N(\widetilde{A}, cz^2 + g)| \leq |c|(\|A\|_{\mathrm{op}} + \|\widetilde{A}\|_{\mathrm{op}})\|A - \widetilde{A}\|_{\mathrm{op}} \\ + \mathrm{Lip}(g)\sqrt{M/N}\|A - \widetilde{A}\|_{\mathrm{op}}. \end{aligned} \tag{6}$$

Indeed each normalized quadratic energy difference is at most $|c|\|A^{\mathsf{T}}A - \widetilde{A}^{\mathsf{T}}\widetilde{A}\|_{\mathrm{op}}$; Cauchy–Schwarz gives the second term uniformly on the input sphere. A uniform energy comparison transfers to logarithms of positive partition functions. Coupling the same singular vectors and using (2) supplies arbitrarily fine finite-block operator approximations with positive limiting masses. Their pressures are therefore Cauchy.

If bounded $g, \widetilde{g}$ have absolute value at most b and $\sup_{|z|\le R} |g(z) - \widetilde{g}(z)| \le \varepsilon$, then

$$|F_N(A, cz^2 + g) - F_N(A, cz^2 + \widetilde{g})| \le (M/N)\varepsilon + 2b\|A\|_{\mathrm{op}}^2/R^2. \tag{7}$$

This counts the field entries outside $[-R, R]$ by their total squared norm. Bounded Lipschitz compact approximation proves the bounded continuous case without changing the quadratic part.

Finally let $g_R = \chi_R g$, where $0 \le \chi_R \le 1$ is continuous, equals one on $[-R, R]$, and vanishes outside $[-2R, 2R]$. With $\epsilon_R = \sup_{|z|>R} |g(z)|/(1 + z^2) \to 0$,

$$|F_N(A, cz^2 + g) - F_N(A, cz^2 + g_R)| \le \epsilon_R(M/N + \|A\|_{\mathrm{op}}^2). \tag{8}$$

Good spectral events make this error small in probability; the same bound controls the finite-block formulas and makes the cutoff limits Cauchy. Conditioning on almost-sure good spectral subsequences handles hypotheses given in probability. Since continuity and subquadratic growth imply $|g(z)| \le C(1 + z^2)$, the absolute pressures are bounded by $C'(M/N + \|A\|_{\mathrm{op}}^2)$. The stated uniform integrability then proves convergence in mean. □

This is an extension of the source's activation class, conditional on its angular theorem, not a new proof of that theorem. In particular, when $c > 0$, substituting $cz^2 + g$ directly into the source's Gaussian terminal can violate its finite-integral domain. Radial extraction avoids that invalid step. The result does not include every continuous function bounded by a quadratic: the same leading coefficient c at both infinities is part of the stated class.

3 Entropy-aware spectral perturbation

In this section g is bounded and Lipschitz, the positive dimension weights v_i are fixed, and zero representatives are interpreted by the continuation above. Write $J_s(\rho)$ for the objective in (5). Since the angular integral is a probability average,

$$\min g \le P_p^g(a) \le \max g.$$

Lemma 3.1 (A physical entropy budget)**.** *Every maximizer of J_s belongs to*

$$\mathcal{K}_{R_s} = \{\rho \in \Delta_I^\circ : D(v\|\rho) \le R_s\}, \qquad R_s = 2\{\alpha \operatorname{osc}(g) + \max_i d_i - v \cdot d\}.$$

A δ-maximizer belongs to $\mathcal{K}_{R_s+2\delta}$.

Proof. The objective is continuous in the interior and tends to $-\infty$ at its boundary, so its supremum is attained. At the reference allocation v its value is at least $v \cdot d + \alpha \min g$. At any allocation it is at most $-D(v\|\rho)/2 + \max d + \alpha \max g$. Comparison proves both assertions. □

For a real vector $e = (e_i)$ and $R \ge 0$ define

$$\Psi_R(e) = \max_{\rho \in \mathcal{K}_R} e \cdot \rho. \tag{9}$$

The following scalar computation is classical convex duality applied to the actual reverse-entropy constraint, not an unweighted norm estimate.

Proposition 3.2 (Exact scalar dual)**.** *If $R = 0$, then $\Psi_R(e) = v \cdot e$. If e is constant, $\Psi_R(e) = e_1$. Otherwise, for $R > 0$, there is a unique $\lambda > \max e$ satisfying*

$$T(\lambda) := \sum_i v_i \log(\lambda - e_i) + \log\left(\sum_i \frac{v_i}{\lambda - e_i}\right) = R. \tag{10}$$

Put $S = \sum_i v_i/(\lambda - e_i)$ and $\eta = 1/S$. Then

$$\rho_i^* = \frac{\eta v_i}{\lambda - e_i}, \qquad \Psi_R(e) = \lambda - \eta. \tag{11}$$

Proof. For fixed $\eta > 0$ the strictly concave function $e \cdot \rho - \eta D(v\|\rho)$ has stationary equations $e_i + \eta v_i/\rho_i = \lambda$. Normalization gives (11). Its entropy equals $T(\lambda)$. If this entropy is R, the same concave objective proves optimality against every allocation with entropy at most R. Writing $S_2 = \sum_i v_i/(\lambda - e_i)^2$, differentiation yields $T' = S - S_2/S < 0$ by strict weighted variance. Also $T(\lambda) \to 0$ as $\lambda \to \infty$. If the total weight at $\max e$ is $m \in (0, 1)$, then $T(\lambda) = (m-1)\log(\lambda - \max e) + O(1) \to \infty$ at the other endpoint. This proves existence and uniqueness, as well as the claimed value after multiplying the stationary equations by ρ_i^*. □

Theorem 3.3 (Block perturbation with entropy control)**.** *Let s, t be two nonnegative spectral representative vectors on the same blocks, with the same v, p, α, c, g. Their structural null representatives, when present, are both zero. Put*

$$R = \max(R_s, R_t), \quad e_i^{(1)} = (s_i - t_i)^2, \quad e_i^{(2)} = |s_i^2 - t_i^2|.$$

Then

$$|\mathcal{F}_{v,s}^{c,g} - \mathcal{F}_{v,t}^{c,g}| \le \sqrt{\alpha}\ \mathrm{Lip}(g)\sqrt{\Psi_R(e^{(1)})} + |c|\Psi_R(e^{(2)}). \tag{12}$$

Proof. Couple uniform directions in the same orthogonal subspaces at two sets of radii. The squared distance of their sum is $M\sum_j(\sqrt{a_j} - \sqrt{a_j'})^2$, independent of the directions. The Lipschitz and Cauchy–Schwarz inequalities therefore give

$$|P_p^g(a) - P_p^g(a')| \le \mathrm{Lip}(g)\sqrt{\sum_j(\sqrt{a_j} - \sqrt{a_j'})^2}.$$

At a common ρ, the resulting change of the angular term is at most $\sqrt{\alpha}\,\mathrm{Lip}(g)\sqrt{\sum_i(s_i - t_i)^2\rho_i}$. The quadratic change is at most $|c|\sum_i |s_i^2 - t_i^2|\rho_i$. Evaluate these bounds first at a maximizer for s, then at a maximizer for t. Lemma 3.1 places both in $\mathcal{K}_R$ and proves the result. □

3.1 What this improves and what it cannot remove

For finite R and nonconstant e, $\Psi_R(e) < \max e$, since all coordinates of an entropy-feasible allocation remain positive. Thus (12) can improve the worst-block operator bound. For example $v = (1/2, 1/2)$, $e = (0, a)$, $a > 0$, and $R = \frac{1}{2}\log(4/3)$ give $\rho_2^* = 3/4$ and $\Psi_R(e) = 3a/4$. This calculation uses the block masses, not the number of displayed coordinates.

It is not a replacement for spectral control by weak convergence alone. For $v = (1 - \varepsilon, \varepsilon)$ and fixed $R > 0$,

$$\Psi_R((0, a)) \longrightarrow a(1 - e^{-R}) \quad (\varepsilon \downarrow 0). \tag{13}$$

Indeed the maximizing second mass $t > \varepsilon$ is the unique upper solution of $(1 - \varepsilon)\log((1 - \varepsilon)/(1 - t)) + \varepsilon\log(\varepsilon/t) = R$. For fixed $t \in (0, 1)$ the left side tends to $-\log(1 - t)$; monotonicity in $t > \varepsilon$ brackets the root and gives (13). A spectrally rare block can still carry a nonvanishing allocation. The source's exceptional-edge examples are therefore not contradicted.

4 A radial-grid enclosure without inverse block masses

Let $m = |I|$, $s_{\max} = \max_i s_i$, $C = \sqrt{\alpha}\,\mathrm{Lip}(g)s_{\max}$, and $D_* = \mathrm{osc}(d)$. For $0 < \epsilon < 1$ put

$$K_\epsilon = \{\rho : \sum_i \rho_i = 1,\ \rho_i \geq \epsilon v_i\}.$$

Theorem 4.1 (Interior and grid error)**.** *Let $\mathcal{N} \subset K_\epsilon$ be a finite ℓ^1 δ-net, and suppose supplied objective values satisfy $|\widehat{J}(\rho) - J_s(\rho)| \leq e$ at every node. Set $Z = \max_{\rho\in\mathcal{N}} \widehat{J}(\rho)$. Then*

$$Z - e \leq \mathcal{F}_{v,s}^{c,g} \leq Z + e + C\sqrt{2\epsilon} + \epsilon D_* + \frac{\delta}{2\epsilon} + C\sqrt{\delta} + \frac{D_*\delta}{2}. \tag{14}$$

No constant here contains $1/\min_i v_i$.

Proof. Replace any interior allocation by $\rho^\epsilon = (1-\epsilon)\rho + \epsilon v$. Concavity of I_v, together with $I_v(v) = 0$ and $I_v(\rho) \leq 0$, shows $I_v(\rho^\epsilon) \geq I_v(\rho)$. The quadratic loss is at most ϵD_*. Moreover

$$\sum_i (\sqrt{\rho_i^\epsilon} - \sqrt{\rho_i})^2 \leq \|\rho^\epsilon - \rho\|_1 \leq 2\epsilon.$$

The previous angular coupling bounds the remaining loss by $C\sqrt{2\epsilon}$. This proves the interior truncation estimate.

On a segment in K_ϵ, the ith entropy derivative is $v_i/(2\rho_i) \leq 1/(2\epsilon)$. Thus entropy changes by at most $\|\rho - \sigma\|_1/(2\epsilon)$. The angular change is at most $C\sqrt{\|\rho - \sigma\|_1}$. Since $\sum_i(\rho_i - \sigma_i) = 0$, subtracting the midpoint of the range of d bounds the linear change by $D_*\|\rho - \sigma\|_1/2$. Move an interior maximizer to a net point and add its oracle error. The lower bound follows from the fact that each node is itself admissible. □

One explicit net is

$$\epsilon v + (1-\epsilon)\{(k_1, \ldots, k_m)/K : k_i \in \mathbb{Z}_{\geq 0},\ \sum_i k_i = K\}.$$

Round down its first $m-1$ coordinates and assign the remaining integer mass to the last coordinate. Its radius is at most $2(m-1)/K$ and its size is $\binom{K+m-1}{m-1}$. For $m = 1$ there is no radial search. For $m > 1$ and a radius bound $0 < \delta < 1$, taking $\epsilon = \delta^{2/3}$ gives an $O(\delta^{1/3}) + e$ enclosure at fixed physical parameters. The dimension cost is real: this is not a dimension-free algorithm. Nor is it a rate in N. In particular, the angular oracle error e has to be justified independently; sampling a finite collection of Gaussian paths does not supply a uniform global oracle.

5 Exact quadratic benchmarks and a dimension obstruction

Proposition 5.1 (Pure quadratic pressure)**.** *For $g = 0$, the finite-block problem has the unique maximizer*

$$\rho_i = \frac{v_i}{2(\lambda - d_i)}, \qquad \lambda > \max d, \qquad \sum_i \frac{v_i}{2(\lambda - d_i)} = 1. \tag{15}$$

Its value is the minimum, over $\lambda > \max d$, of

$$\lambda - \frac{1}{2} - \frac{1}{2}\sum_i v_i \log(2(\lambda - d_i)).$$

Proof. Strict concavity of $I_v + d\cdot\rho$ and its divergent boundary cost give an interior unique maximum. Its stationary equations give (15); the sum decreases from infinity to zero as λ runs from $\max d$ to infinity. At the solution, $d\cdot\rho = \lambda - 1/2$. Substitution gives the value. The displayed scalar objective has derivative $1 - \sum_i v_i/[2(\lambda - d_i)]$ and positive second derivative, proving the minimum statement. Alternatively $\log u \le u - 1$ gives the primal–dual inequality directly. □

This is a classical spherical-integral calculation, not an additional novelty claim. It supplies a stringent nonlinear check on normalizations. For $\alpha \ge 1$ and $A^{\mathsf{T}}A = s^2 I$, every finite-N pressure is exactly cs^2. For $\alpha < 1$ with all visible singular values equal to s, put ρ for the visible input fraction. The maximizing equation is

$$\alpha - \rho + 2cs^2\rho(1-\rho) = 0.$$

At $\alpha = 1/2$ and $cs^2 = 2/3$, it gives $\rho = 3/4$ and $F = 1/2 + \frac{1}{4}\log(3/4)$. Replacing v by output proportions, or omitting the input-null entropy, would give a different answer.

Proposition 5.2 (Why the dimension convention matters)**.** *Theorem 2.1 cannot in general be extended to arbitrary $M/N \to 1$ with the same limiting formula and only singular-slot control.*

Proof. Take $f(z) = -z^2$ and all rectangular singular values equal to one. Choose independent Haar singular bases, so the model is bi-orthogonally invariant and its slot law is exactly δ_1, with no outlier. For $M = N$, the pressure is -1. For $M = N - 1$, the visible squared fraction has law $V \sim \mathrm{Beta}((N-1)/2, 1/2)$, and

$$F_N = \frac{1}{N}\log \mathbb{E}e^{-NV} \longrightarrow \sup_{0<q\le 1}\{\tfrac{1}{2}\log q - q\} = -\frac{1}{2} - \frac{1}{2}\log 2 > -1.$$

The beta normalization has logarithm $o(N)$. On compact subintervals of $(0,1)$ its extensive density is $\exp\{(N/2)\log q + o(N)\}$. Near zero the entropy tends to $-\infty$; the integrable endpoint factor $(1-q)^{-1/2}$ contributes no extensive rate near one. These facts prove both Laplace bounds. The maximizer is $q = 1/2$. Alternating the two dimension choices even destroys convergence. □

The source's appendix proves a different assertion for *bounded* activations by deleting two bounded output terms and comparing balls. For $-z^2$ those deleted terms are not bounded uniformly on the sphere. Thus this obstruction does not refute that source assertion. It corrects a tempting overextension of the new quadratic theorem. The same example also makes the distinction between a structural null direction and a zero inside the controlled singular slots indispensable.

6 Normalized nonlinear expectation and original comparisons

For a fixed probability law γ, bounded real U, h, and $0 < \zeta \le 1$, set

$$T_\zeta(U) = \zeta^{-1}\log\int e^{\zeta U}\,d\gamma, \qquad d\gamma_U = \frac{e^{\zeta U}d\gamma}{\int e^{\zeta U}d\gamma}.$$

Direct differentiation under the bounded integrals gives

$$DT_\zeta(U)h = \int h\,d\gamma_U, \quad D^2T_\zeta(U)[h,h] = \zeta\,\mathrm{Var}_{\gamma_U}(h) \le \frac{\zeta}{4}\,\mathrm{osc}(h)^2. \tag{16}$$

The last inequality follows from $(h - \min h)(\max h - h) \ge 0$ and completing the square in its mean. In particular

$$T_\zeta(U) + \mathbb{E}_{\gamma_U}h \le T_\zeta(U+h) \le T_\zeta(U) + \mathbb{E}_{\gamma_U}h + \frac{\zeta}{8}\,\mathrm{osc}(h)^2.$$

The full normalizing denominator matters; replacing the tilted mean by an untilted one discards the first-order response. Also $|T_\zeta(U) - T_\zeta(V)| \le \|U - V\|_\infty$ and $T_\zeta(U + a) = T_\zeta(U) + a$. Successive nested expectation steps therefore accumulate uniformly certified step errors additively. There is no extra factor $1/\zeta$ in this sup-norm error propagation. These are classical baseline facts, not a new Gaussian hierarchy theorem. For the source's unbounded negative quadratic penalties, a bounded-domain calculation still needs the source's separate Gaussian row-tail control. Errors known only at finitely many sampled arguments do not qualify as uniform step errors.

Three freshly read private originals provide comparisons, not missing probabilistic theorems. The physical-metrics chapter [6] derives the induced Gram matrix $G = J^*J$, the physical adjoint, and the decomposition into resolved and unresolved residual energy. Here that principle requires keeping $v_j = \alpha p_j$, the input radius N, and the probability normalization; it does not produce Haar majorization. The local-moments chapter [7] distinguishes exact polynomial moments from Taylor remainders and from an actual closure theorem. Here the quadratic energy is exactly constant on a radial fiber, whereas $e^{g(z)}$ does not become polynomial merely because g is approximated by a polynomial. Finally the original nonlinear-integral report [8] differentiates a normalized kernel, including its quotient correction. That is the relevant algebraic analogue of (16), not evidence for matrix synchronization or the cavity lower bound.

7 Scope, costs, and current public context

The proved increments are the conditional growth extension, the entropy-localized block comparison, and the explicit radial enclosure. The dimension obstruction is also proved locally. Classical convex duality, Cauchy–Schwarz, Dirichlet large deviations, and exponential tilting are credited as tools, not renamed discoveries. The source's deep Gaussian and Haar mechanisms were read but are not independently certified by this report.

The advantages are quantitative and specific: the quadratic term has an exact physical coefficient, block perturbations can use an entropy budget rather than the largest block error, and radial truncation has no inverse-smallest-mass constant. The costs are equally concrete: the scalar dual needs certified logarithms for a numerical enclosure; the grid cardinality grows rapidly with the number of blocks; bounded Lipschitz constants or explicit approximation errors are required; and the angular values remain nontrivial input data. These statements give no convergence rate for the source's ordered witness limits.

Primary public records were refreshed on October 9, 2026, using only public titles and URLs. Kabashima [2] and Shinzato–Kabashima [3] are direct spectral-model antecedents; the latter's abstract describes replica and TAP calculations. Montanari–Zhou [4] distinguish a nonrigorous Parisi prediction from their rigorous algorithmically realizable projection class. Guionnet–Maida [5] provide a classical spherical-integral context. Only the primary abstracts and metadata of these external works were inspected, not their full proofs. None is being used as independent confirmation of every hypothesis or conclusion of the cached source. The cached source was byte-authenticated against the captured October 8 tree; this is provenance, not a live-branch or proof-status assertion.

Useful next steps would be a rigorous angular oracle with certified uniform error, an entropy-aware adaptive choice of spectral blocks, and a separate theorem handling subextensive structural null spaces for quadratic growth. The explicit counterexample shows that the last task must retain extra geometry or an exceptional-allocation variable. No general activation beyond the stated quadratic-plus-subquadratic class is asserted.

8 Verification boundaries

The owned read ledger records the full 1,465-line public manuscript, its one-line wrapper, 47-line glyph mapping, and 18-line README: 1,531 public lines in total. Fresh private literal reads cover 210 lines of the physical-metrics chapter, lines 1–240 of the 446-line local-moments chapter, and lines 335–453 of the 2,250-line nonlinear-integral report. Their unread complements, embedded inputs not opened, and pre/post hashes are explicit. This is 569 private lines, not an exhaustive corpus read. The last private interval ends inside a later equation; only its completed earlier proof passages are used here.

Owned exact rational tests accompany the local algebra, entropy dual, allocation comparison, and radial grid: 13,975 parameter cases and 47,428 individual assertions passed. Of these, 3,963 entropy-feasible comparisons encode logarithmic constraints as rational products, so no floating-point logarithm is used to decide feasibility. They test finite fixtures, not random-matrix limits, synchronization, Gaussian tail uniformity, or the full Haar formula. Parent checks, if any, are separate and are not included in the owned counts. The final audit records the exact case and assertion totals, native compiler outcome, exported PDF hash, and individual page inspections.

References

[1] *The spherical perceptron with bi-orthogonally invariant disorder*, September 24, 2026, cached manuscript, 1,465 lines. Public source directory. The audit records the pinned byte identity.

[2] Y. Kabashima, *Inference from correlated patterns: a unified theory for perceptron learning and linear vector channels*, J. Phys.: Conf. Ser. 95 (2008), 012001. Primary preprint record.

[3] T. Shinzato and Y. Kabashima, *Perceptron capacity revisited: classification ability for correlated patterns*, J. Phys. A 41 (2008), 324013. Primary preprint record.

[4] A. Montanari and K. Zhou, *Which exceptional low-dimensional projections of a Gaussian point cloud can be found in polynomial time?*, arXiv:2406.02970, version 3, October 17, 2025. Primary preprint record.

[5] A. Guionnet and M. Maida, *A Fourier view on the R-transform and related asymptotics of spherical integrals*, J. Funct. Anal. 222 (2005), 435–490. Publisher record; associated preprint arXiv:math/0406121, titled *An asymptotic log-Fourier interpretation of the R-transform.*

[6] Private Ram corpus, volume 21, *Physical metrics* chapter. Fresh read: lines 1–210. Confidential local comparison; not uploaded.

[7] Private Ram corpus, revised volume 1, *Local moments* chapter. Fresh read: lines 1–240; later lines and subsidiary inputs not read.

[8] Private Ram nonlinear integral-equation/PDE corpus, original TR6. Fresh read: lines 335–453; normalized-kernel proof at 399–425. Confidential local comparison, not a global spin-glass input.